

Section 1: Critical Thinking, Research and Analytical Skills Testing

As a data analyst supporting the research team in curating data and insights to back up their research on civic technology, please perform the following task;

Highlight the steps you will take to curate credible secondary datasets that support the research work.

1. **Understand the research objective** in order to narrow down on the specific questions the team is addressing in civic technology. Trying to understand what exactly is the kind of data that is required for this research? In civic technology, data that could be used includes: data on public services, government transparency, citizen engagement, public service delivery, infrastructure, socio-economic indicators etc.
2. **Identify relevant data sources** which may include:
 - Publicly available datasets from government databases e.g. open data portals (like from Kenya National Bureau of Statistics, KNBS), census data (also from KNBS), etc.
 - Data from international organizations like the UN and World Bank. They can provide information on governance, civic engagement and economic information about different countries/regions.
 - Datasets that have been published by NGOs and research institutions that focus on civic matters.
 - Data shared by some tech companies and civic tech platforms.
3. **Ensure that the dataset comes from a reputable and trusted source**, like from the above sources, KNBS, UN and World Bank can be considered as reputable sources. I would also check to make sure that the data is up-to-date, or is at least **representative of the time period in question**, and that it **covers the geographical region(s) of interest**.
4. Document the dataset information, i.e. explain meaning of variables and any other key information related to the dataset. I would also clearly reference the original source of data to maintain transparency.
5. Prepare the data for analysis by cleaning it first. This would involve handling missing or inconsistent data, and transforming the data into usable format for analysis.

Analysis of the data

Based on the crime trends data for the years 2008 to 2017, some actionable insights, that can inform decisions related to public safety, policy-making, and resource allocation, can be drawn.

1. Consistent increase in breakings

Breakings (burglaries, thefts from homes, etc.) have the highest and most consistent number of reported cases over the years.

- This indicates a growing concern related to personal and property safety.
- The increase in breakings could be attributed to factors like poverty, unemployment, or social inequality, but it also suggests the need for better community policing and crime prevention initiatives in areas most affected by property-related crimes.

Actionable insights

- Increase funding for local law enforcement efforts in high-crime areas to prevent burglaries.
- Implement or expand community-based programs such as neighborhood watch groups, crime reporting platforms, and youth engagement initiatives to reduce opportunities for break-ins.
- Implement strict penalties for police officers who ignore criminal activities in exchange for bribes or kickbacks.

2. Fluctuations in robbery cases

Robbery cases exhibit some fluctuations, with increases and decreases over the years.

- The increase in robbery cases in years like 2009 and 2011 suggests that economic factors, such as unemployment or income disparity, might be linked to the rise in violent crime.

Actionable insights

- Start campaigns to fight robbery, teach people how to stay safe, and share tips on protecting their home, like installing strong locks, using security lights, and reporting suspicious activity.
- Improve safety by increasing patrols in high-crime areas and using past data to predict spots where robberies might happen before they do.

3. Economic crimes on the rise

Economic crimes show an increasing trend, especially starting from 2010, and have consistently been on the rise since then.

- This could indicate a growing concern in areas such as fraud, corruption, and financial crimes.

Actionable insights

- Improve financial transparency by making government budgets and spending public, using digital systems to track money flow, and ensuring regular audits.
- Support whistleblowers and stronger public oversight to fight corruption.
- Run public awareness campaigns to teach people about economic crimes, how to avoid fraud, and their rights as consumers, to help reduce financial misconduct.

4. Relatively low crime in traffic offenses

Traffic offenses consistently show low numbers compared to other crimes, with relatively minor fluctuations over the years.

- This may suggest that traffic law enforcement and road safety programs receive less focus or generate fewer reported incidents compared to more violent or high-profile crimes.

Actionable insights

- Although traffic offenses are generally low, we can consider analyzing location-specific trends to identify hot spots. Target these areas with road safety awareness campaigns focused on common issues like speeding and drunk driving, especially in regions with slightly higher traffic-related offenses.

Conclusion

Based on the trends observed in the dataset, the following actionable steps can be recommended:

1. Increase law enforcement presence and community engagement in areas with high property crime and robbery.
2. Focus on anti-corruption initiatives and transparency programs to fight the growing rise of economic crimes.
3. Strategic resource allocation in order to prioritize areas where the most significant crime spikes occur.
4. Targeted crime prevention campaigns to address specific crime categories like theft, robbery, and economic crimes.